

Algebraic Number Theory: Assignment

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Some Examples

Let R be an integral domain with fraction field F . Let K be a field extension F and let V be a finite dimensional K -vector space.

1. L is an R -submodule of $F \otimes_R L$. To analysis, we can consider

$$(v_1, \dots, v_m) = (u_1, \dots, u_n)P$$

Where (u_i) and (v_i) is the basis of $F \otimes_R L$ and generators of L . We can also extend $v_1, \dots, v_m$ to a basis of $F \otimes_R L$ by some consideration.

2. An example of R -lattice not support in V . Let $K = V = \mathbb{R}$ and $R = \mathbb{Z}$. Then $L = \mathbb{Z} + \mathbb{Z}\sqrt{2}$ is a $\mathbb{Z}$ -lattice but not support in $\mathbb{R}$.

3. It is easy to see that from 2., $\dim L \geq \dim_K \text{Span}_K L$ and a system of vectors in L is linearly independent over R iff. over K .

4. We can clearly see from 2. the equivalence of existence of K -free basis and support in V .

5. Let $L = 2 \cdot \mathbb{Z} + \mathbb{Z}(1 + \sqrt{-5})$, $K = \mathbb{Q}(\sqrt{2})$ and $V = \mathbb{Q}(\sqrt{2}, \sqrt{-5})$. Then L is support in V . L is a non-trivial generator of ideal class group of $\mathbb{Q}(\sqrt{-5})$.

6. Let $L = \mathbb{Z} + \mathbb{Z}\sqrt{-5}$ and $L' = 2 \cdot \mathbb{Z} + \mathbb{Z}(1 + \sqrt{-5})$. Then $\det(L/L') = 2 \cdot \mathbb{Z}^*$ and $2L \subseteq L'$.

Norm and discriminant in Algebraic Number Theory

We consider L/K is a separable finite algebraic extension. We specifically let O_K is not a PID. Let I be a fraction ideal of O_L . Then there exists a basis $\{\omega_i\}$ of L over K such that

$$I = \mathfrak{a}_1\omega_1 \oplus \mathfrak{a}_2\omega_2 \oplus \dots \oplus \mathfrak{a}_n\omega_n.$$

Similarly, we have

$$\mathcal{O}_L = \mathfrak{b}_1\nu_1 \oplus \mathfrak{b}_2\nu_2 \oplus \dots \oplus \mathfrak{b}_n\nu_n.$$

Suppose that P is the transform matrix such that $(\omega_i) = P(\nu_i)$. In this sense, the norm of I is given by

$$\det(P) \cdot \frac{\mathfrak{a}_1 \dots \mathfrak{a}_n}{\mathfrak{b}_1 \dots \mathfrak{b}_n}$$

The discriminant of I is given by

$$(\mathfrak{a}_1\mathfrak{a}_2 \dots \mathfrak{a}_n)^2 \cdot \det(\text{Tr}_{L/K}(\omega_i\omega_j)).$$

The discriminant of field L is the discriminant of O_L . If $L = K(\theta)$ with minimal polynomial $p(x)$ then the discriminant $\text{disc}(f)$ of f is given by the discriminant of lattice generated by $\{\theta^n\}$.

7. Let $L = \mathbb{Z} + \mathbb{Z}\sqrt{-5}$ and $L' = \mathbb{Z}(3 - \sqrt{-5}) + \mathbb{Z}(9 + \sqrt{-5})$. Let P be the transition matrix. Here is the triangular process of P . Let $L'^{(1)} = 3\mathbb{Z}$ and $L'^{(2)} = L$. Then $\pi_1(L'^{(1)}) = 3\mathbb{Z}$ and $\pi_2(L'^{(2)}) = \sqrt{-5}\mathbb{Z}$ is principal. Let $d = \det(L/L') = 12$. Then $12\mathbb{Z} \subseteq 3\mathbb{Z}$ and $12\mathbb{Z}\sqrt{-5} \subseteq \mathbb{Z}\sqrt{-5}$ it implies that $\pi_i(L'^{(i)})$ is not zero. Choose $f_i \in L'^{(i)}$ such that $\pi_i(f_i) = t_{ii}$ where t_{ii} is the generator of ideal $\pi_i(L'^{(i)})$.

The core of process above is the coefficient ideal $\pi_i(L'^{(i)})$. And the condition $L' \subseteq L$ guarantee the projection is not zero ideal.

8. Resay a trick. If we don't have $L' \subseteq L$, we just need to consider $dL' \subseteq L$.

We consider field extension L/K equipped with bilinear form $\text{Tr}_{L/K}$.

9. If L/K is separable then Tr is not singular.

10. If L/K is inseparable extension, then Tr is totally singular. Let $L = \mathbb{F}_p(t^{1/p})$ and $K = \mathbb{F}_p(t)$ then $\text{Tr}(L) = p$.

11. Isotropic case: Let $L = \mathbb{Q}(i)$ and $L = \mathbb{Q}$. Consider $\text{Tr}((1+i)^2) = 0$.

12. Anisotropic case: Let $L = \mathbb{Q}(\sqrt{2})$ and $L = \mathbb{Q}$. Consider $\text{Tr}(x^2) = x^2 + \bar{x}^2 \neq 0$ if $x \neq 0$.

13. Alternative case: Twisted Trace. Let $L = \mathbb{Q}(i)$ and $L = \mathbb{Q}$ with $B(x, y) = \text{Tr}(i \cdot \bar{x}y)$.

Twisted Trace Form and Different

Let L/K be a finite separable algebraic extension. The Different of L/K is defined by $(*\mathcal{O}_L)^{-1}$.

14. Consider vector space $\mathbb{F}_2x$ with quadratic form $Q(x) = 1$. Then radical is $\mathbb{F}_2x$ but it is not singular.

A.1 Homework 1

Throughout what follows, let R denote an integral domain with fraction field K .

Exercise A.1.1.

Suppose R is a local domain. Let I be a fractional ideal of R . Prove that I is invertible if and only if it is principal.

Hint: There exist $x \in I$, $y \in I^{-1}$ such that $xy \in R^\times$. Then prove $I = xR$.

Solution: Suppose $x \neq 0$ and $I = xR$ is principal then let $J = x^{-1}R$ and $IJ = R$ implies I is invertible. Conversely, if there exist a fractional ideal J such that $IJ = R$. Then there exist $x_i \in I$ and $y_i \in J$ such that

$$\sum x_i y_i = 1.$$

Because R is local, without loss of generality, we can suppose $x_1 y_1 \in R \setminus \mathfrak{m} = R^\times$. Since $x_1 \in I$, $x_1 R \subseteq I$ is obvious. Hence $y_1 \in J$ implies $y_1 I \subseteq R$, multiplying by x_1 on both sides, we obtain that $I \subseteq x_1 R$. Therefore, $I = x_1 R$ is principal. ■

Exercise A.1.2.

Prove that a Dedekind domain with only finitely many prime ideals is a PID.

Solution: Let $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ be all prime ideals of Dedekind domain R . It suffices to prove all prime ideals are principal. Without loss of generality, we prove $\mathfrak{p}_1$ is principal and others are in the same way. Let $x \in \mathfrak{p}_1 \setminus \mathfrak{p}_1^2$ and by Chinese Remainder Theorem, there exist $u \in R$ such that

$$u \equiv x \pmod{\mathfrak{p}_1^2} \text{ and } u \equiv 1 \pmod{\mathfrak{p}_i} \text{ for } i \neq 1.$$

Let $v_1, \dots, v_n$ be the valuations of local rings $R_{\mathfrak{p}_1}, \dots, R_{\mathfrak{p}_n}$ respectively. Then we obtain that

$$v_1(u) = 1 \text{ and } v_i(u) = 0 \text{ for } i \neq 1,$$

because $u - x \in \mathfrak{p}_1^2$ and implies $u \in (x) + \mathfrak{p}_1^2 \subseteq \mathfrak{p}_1$ meaning that $v_1(u) \geq 1$. Suppose $v_1(u) \geq 2$, equivalently $u \in \mathfrak{p}_1^2$, then $x \in (u) + \mathfrak{p}_1^2 = \mathfrak{p}_1^2$ contradicting our assumption, therefore $v_1(u) = 1$. Because R is Dedekind domain, the unique factorization of fractional ideals implies

$$(u) = \mathfrak{p}_1^{v_1(u)} \dots \mathfrak{p}_n^{v_n(u)} = \mathfrak{p}_1,$$

the principality of $\mathfrak{p}_1$ follows. ■

Exercise A.1.3.

Suppose R is a DVR with maximal ideal $\mathfrak{p}$. Let $k = R/\mathfrak{p}$ and $U = R^\times$. Let v be the normalized discrete valuation of K associated to R . For each $n \geq 1$, define

$$U_n = 1 + \mathfrak{p}^n = \{x \in K \mid v(x - 1) \geq n\}.$$

1. Show that each U_n is a subgroup of U and that $\bigcap_{n=1}^{\infty} U_n = 1$.
2. Prove that the natural quotient map $R \rightarrow k = R/\mathfrak{p}$ induces an isomorphism of multiplicative groups $U/U_1 \cong k^*$.
3. Prove that for each $n \in \mathbb{Z}$, there is an isomorphism of R -modules $k = R/\mathfrak{p} \xrightarrow{\sim} \mathfrak{p}^n/\mathfrak{p}^{n+1}$.
4. Prove that for each $n \geq 1$, the map $u \mapsto u - 1$ induces an isomorphism of abelian groups $U_n/U_{n+1} \xrightarrow{\sim} \mathfrak{p}^n/\mathfrak{p}^{n+1}$.
5. Let p be the characteristic of k . (So $p = 0$ or p is a prime number.)
 - (a) Show that $U_n^p := \{a^p \mid a \in U_n\}$ is contained in U_{n+1} for every $n \geq 1$.
 - (b) Let $m \geq 1$ be an integer not divisible by p . Prove that for every $n \geq 1$, the group homomorphism

$$U_n \longrightarrow U_n ; \quad u \longmapsto u^m$$

is injective.

Solution: (1) Closed under multiplication holds since $(1 + \mathfrak{p}^n)(1 + \mathfrak{p}^n) \subseteq 1 + \mathfrak{p}^n$. Identity element is 1. Suppose $x \in U_n$, then $v(x^{-1} - 1) = v(x^{-1}) + v(1 - x)$ hence x is invertible meaning that $v(x) = 0$ and $v(1 - x) = v(x - 1)$ implies $v(x^{-1} - 1) \geq n$. Suppose $x \in \bigcap_{n=1}^{\infty} U_n$, then we obtain $v(x - 1) = \infty$, equivalently, $x = 1$. (2) The natural quotient map restricted to U , denoted by $\pi : U \rightarrow k^*$, induces isomorphism $U/U_1 \cong k^*$ because $\ker(\pi) = U_1$. (3) Consider commutative graph

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathfrak{p} & \hookrightarrow & R & \twoheadrightarrow & k \longrightarrow 0 \\ & & \downarrow \cdot u^n \cong & & \downarrow \cdot u^n \cong & & \downarrow \cong \bar{\phi} \\ 0 & \longrightarrow & \mathfrak{p}^{n+1} & \hookrightarrow & \mathfrak{p}^n & \twoheadrightarrow & \mathfrak{p}^n/\mathfrak{p}^{n+1} \longrightarrow 0 \end{array}$$

which the third isomorphism follows from the Five Lemma. (4) Consider a surjective group homomorphism

$$\varphi : U_n \rightarrow \mathfrak{p}^n/\mathfrak{p}^{n+1} \quad u \mapsto u - 1 + \mathfrak{p}^{n+1}$$

with kernel U_{n+1} , then the first isomorphism theorem yields isomorphism $U_n/U_{n+1} \cong \mathfrak{p}^n/\mathfrak{p}^{n+1}$. (5.a) For $p = 0$, obvious to see that $U_n^0 = \{1\} \subseteq U_{n+1}$. For $p \neq 0$ and any $a \in U_n$,

$$v(a^p - 1) = v((a - 1)^p) = p \cdot v(a - 1) \geq n + 1$$

implies $a^p \in U_{n+1}$. (5.b) It suffices to prove for $m = q$ where q is a prime number and coprime to p . Suppose that $u \in U_n$ such that $u^q = 1$. According to Bézout's identity and (5.a), suppose $ap + bq = 1$, then $u = u^{ap+bq} = u^{ap} \in U_{n+1}$. Thus we can consider a new group homomorphism: $U_{n+1} \rightarrow U_{n+1}$. Continuing this process, we obtain that $u \in \bigcap_{n=1}^{\infty} U_n$, then $u = 1$ and injectivity holds. ■

Exercise A.1.4.

Prove that if a domain R is integrally closed, then so is the polynomial ring $R[t]$.

Solution: Let $f = p/q$ be an arbitrary integral element in $R(t)$. Without loss of generality, we can suppose q is irreducible, because if $q = q_1 q_2$ then we consider integral element $f q_1$, and suppose p, q are coprime. Suppose f satisfies a monic polynomial $f^n + a_{n-1}f^{n-1} + \dots + a_0 = 0$ where $a_k \in R[t]$. Then, by multiplying both side by q^n , we obtain that $p^n + a_{n-1}qp^{n-1} + \dots + a_0q^n = 0$, which implies that $q|p^n$. Let $F = \text{Frac}(R)$ be the fraction field of R , and interpret p, q as elements in $F[t]$, then we obtain that $q|p$ since $F[t]$ is an UFD and hence $q \in F^*$ because p, q coprime. Therefore, we can write $f = c_m t^m + \dots + c_0$ where $c_k \in F$. Let $f_1 = t^r - f$, where $r = \deg f + \sum \deg a_k$, then f_1 is the root of polynomial

$$P(X) = (t^r - X)^n + a_{n-1}(t^r - X)^{n-1} + \dots + a_0.$$

Let $X = 0$, we obtain the constant term of this polynomial

$$t^{rn} + a_{n-1}t^{rn-r} + a_{n-2}t^{rn-2r} + \dots + a_0$$

denoted by $P(0)$. Let $P(X) = XQ(X) + P(0)$, then from $P(f_1) = 0$ we obtain that

$$P(0) = -f_1Q(f_1),$$

the roots of f_1 are among the roots of monic polynomial $P(0) \in R[t]$, the coefficients of f as of f_1 are elementary symmetric polynomials of its roots, which are integral over R . Therefore, we obtain that the coefficients of f are contained in R , $R[t]$ is integrally closed. ■

Exercise A.1.5.

Let $A = \mathbb{Q}[X, Y]/(X^2 - Y^3)$. Show that A is a domain but it is not integrally closed.

Solution: $(X^2 - Y^3)$ is a prime ideal because $X^2 - Y^3$ is irreducible and $\mathbb{Q}[X, Y]$ is UFD, *a fortiori*, A is a domain. Hence, X/Y is the root of a monic polynomial $P(Z) = Z^2 - Y$ but not contained in A . If it is contained, then there exist $f, g \in \mathbb{Q}[X, Y]$ such that

$$X = Yf - (X^2 - Y^3)g$$

implies contradiction by comparing their linear terms. ■

Exercise A.1.6.

Let R be a Dedekind domain and let $\mathfrak{a} \subseteq R$ be a nonzero ideal. Show that the quotient ring $R/\mathfrak{a}$ has only finitely many ideals and that all of these ideals are principal.

Solution: By isomorphism theorem of rings, the natural quotient $R \rightarrow R/\mathfrak{a}$ gives one-to-one correspondence of the ideals of R containing $\mathfrak{a}$ and the ideals of $R/\mathfrak{a}$. Therefore, all ideals of $R/\mathfrak{a}$ can be written as the form

$$\mathfrak{p}_1^{i_1} \dots \mathfrak{p}_n^{i_n} + \mathfrak{a}$$

where $\mathfrak{p}_k$ are the prime ideals of R containing $\mathfrak{a}$ and $0 \leq i_k \leq v_k(\mathfrak{a})$ where v_k is the valuation associated to $\mathfrak{p}_k$. The principality directly from *Exercise A.1.2*. ■

Exercise A.1.7.

Show that every integral ideal of a Dedekind domain can be generated by two elements.

Solution: Let $\mathfrak{a}$ is an arbitrary ideal of R with prime ideal decomposition

$$\mathfrak{a} = \mathfrak{p}^k \mathfrak{b}$$

Where $\mathfrak{p}$ is a prime ideal coprime to the ideal $\mathfrak{b}$. The quotient ring $R/\mathfrak{p}^{k+1}\mathfrak{b}$ is principal from *Exercise A.1.6*. Therefore we can assume that $\mathfrak{a} = (a) + \mathfrak{p}^{k+1}\mathfrak{b}$. Hence, consider another quotient ring $R/(a)$, we obtain that $\mathfrak{a} = (a, b)$. ■

A.2 Homework 2

Exercise A.2.1

Let R be the ring of algebraic integers in $\mathbb{C}$, i.e., the integral closure of $\mathbb{Z}$ in $\mathbb{C}$. Consider the ideal $I \subseteq R$ generated by the infinite subset $\{\sqrt[n]{2} \mid n \in \mathbb{N}\}$.

Show that I is not finitely generated. Deduce that R is not Noetherian (and hence not a Dedekind domain).

Solution: Suppose that I is finitely generated with generators $\alpha_1, \dots, \alpha_m$. According to Chevalley's Extension Theorem for Valuations, we can extend 2-adic valuation on $\mathbb{Q}$ to $\overline{\mathbb{Q}}$, let v_2 denote this extended valuation. By definition, we obtain that $R \subseteq \overline{\mathbb{Q}}$ and $v_2(R) \geq 0$. Since I is generated by $\{\sqrt[n]{2}\}$ and hence $v_2(\sqrt[n]{2}) = 2^{-n}$ is strictly greater than zero, the valuation $v_2(\alpha_i)$ is strictly greater than zero. Let $\varepsilon = \min\{v_2(\alpha_i)\}$ be the lower bound of valuations of finite generators α_i . For arbitrary $r \in I$, we can write $r = \sum_{i=1}^m c_i \alpha_i$ for some $c_i \in R$. Using the non-Archimedean property and $v_2(c_i) \geq 0$, we obtain:

$$v_2(r) \geq \min_i \{v_2(c_i) + v_2(\alpha_i)\} \geq \min_i \{v_2(\alpha_i)\} = \varepsilon > 0$$

However we find that $v_2(\sqrt[n]{2}) = 2^{-n}$ converging to zero, which is a contradiction. ■

Exercise A.2.2

Let R denote a Dedekind domain with fraction field K and suppose $R \neq K$.

Prove that there exists a set Ω of normalized discrete valuations of K such that the following two properties hold:

- (R1) $R = \{x \in K \mid v(x) \geq 0 \text{ for all } v \in \Omega\}$;
- (R2) for every $a \in K^\times$, the set $\{v \in \Omega \mid v(a) \neq 0\}$ is finite.

Solution: Let

$$\Omega = \{v_{\mathfrak{p}} \mid \mathfrak{p} \in \text{Spm}(R)\}$$

where $v_{\mathfrak{p}}$ is the normalized discrete valuation associated with $R_{\mathfrak{p}}$. For (R1), we find that $v_{\mathfrak{p}}(x) \geq 0$ if and only if $x \in R_{\mathfrak{p}}$, then we obtain that

$$\{x \in K \mid v(x) \geq 0 \text{ for all } v \in \Omega\} = \bigcap_{\mathfrak{p} \in \text{Spm}(R)} R_{\mathfrak{p}} = R.$$

For (R2), we first fix an arbitrary element $a \in K^*$ and let A denote the set $\{v \in \Omega \mid v(a) \neq 0\}$. Since R is Dedekind, we may suppose that

$$aR = \mathfrak{p}_1^{a_1} \dots \mathfrak{p}_n^{a_n}.$$

where $a_i \neq 0$. I claim that $A = \{v_{\mathfrak{p}_1}, \dots, v_{\mathfrak{p}_n}\}$. For an arbitrary maximal ideal $\mathfrak{q} \notin \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$, it does not appear in the factorization of aR , yielding $v_{\mathfrak{q}}(a) = 0$ and hence $v_{\mathfrak{q}} \notin A$. This implies $A \subseteq \{v_{\mathfrak{p}_1}, \dots, v_{\mathfrak{p}_n}\}$. Conversely, by definition we have $v_{\mathfrak{p}_i}(a) = a_i \neq 0$, meaning $\{v_{\mathfrak{p}_1}, \dots, v_{\mathfrak{p}_n}\} \subseteq A$. Therefore, A is precisely this finite set. ■

Exercise A.2.3

Let R denote an integral domain with fraction field K . Let Ω be a set of normalized discrete valuations of K satisfying Properties (R1) and (R2) in Exercise A.2.2. For each $v \in \Omega$, let R_v be its valuation ring and let $\mathfrak{p}_v$ be the maximal ideal of R_v .

Suppose further that every nonzero prime ideal of R is maximal.

1. Let $\mathfrak{p}$ be a maximal ideal of R .
 - (a) Let $z \in K^\times$ be such that $R \cap z^{-1}R \subseteq \mathfrak{p}$. Show that there exists $v \in \Omega$ such that $\mathfrak{p} = R \cap \mathfrak{p}_v$ and $v(z) < 0$.
 - (b) Prove that $\Omega_{\mathfrak{p}} := \{v \in \Omega \mid R_v \supseteq R_{\mathfrak{p}}\}$ is a finite set.
 - (c) Prove that $R_{\mathfrak{p}} = R_v$ for a unique $v \in \Omega$. Thus $\Omega_{\mathfrak{p}}$ consists of a single element.
2. Prove that for every $v \in \Omega$, there exists a unique maximal ideal $\mathfrak{p}$ such that $R_v = R_{\mathfrak{p}}$.
3. Deduce from (1) and (2) that the map

$$\Omega \longrightarrow \text{Spm}(R) ; v \longmapsto \mathfrak{p}_v \cap R$$

is well defined and bijective. Let $\mathfrak{p} \mapsto v_{\mathfrak{p}}$ denote its inverse map.

4. Let $I \subseteq R$ be a nonzero ideal.
 - (a) Prove that $\{\mathfrak{p} \in \text{Spm}(R) \mid v_{\mathfrak{p}}(I) \neq 0\}$ is finite, where $v_{\mathfrak{p}}(I) := \min\{v_{\mathfrak{p}}(x) \mid x \in I\}$.
 - (b) Prove that there is a unique factorization $I = \prod_{\mathfrak{p} \in \text{Spm}(R)} \mathfrak{p}^{r_{\mathfrak{p}}}$, where each $r_{\mathfrak{p}} \in \mathbb{N}$ and almost all of them are zero.
 - (c) If $J \subseteq R$ is another nonzero ideal, show that $I \mid J$ if and only if $I \supseteq J$.
5. Prove that R is a Dedekind domain.

Solution: (1.a) First I claim that there exists $v \in \Omega$ such that $v(z) < 0$. If not, then for every $v \in \Omega$ we have $v(z) \geq 0$. In this case, we have $v(zR_z) \geq 0$ and therefore

$$R_v \subseteq z^{-1}R_v.$$

Hence we find that

$$\mathfrak{p} \supseteq R \cap z^{-1}R = R \cap \left(\bigcap_{v \in \Omega} z^{-1}R_v \right) \supseteq R \cap \left(\bigcap_{v \in \Omega} R_v \right) = R$$

which is a contradiction. Let

$$A = \{v \in \Omega \mid v(z) < 0\}.$$

By (R2), A is a finite set, then we say $A = \{v_1, \dots, v_n\}$. We suppose that $v_i(z) = -a_i$, then we find that

$$R \cap z^{-1}R = \{x \in K \mid v_i(x) \geq a_i, v(x) \geq 0 \text{ for any } v \in \Omega\}.$$

And therefore

$$R \cap z^{-1}R = \left(\bigcap_{v_i \in A} R \cap \mathfrak{p}_{v_i}^{a_i} \right).$$

Since the product of ideals is always contained in their intersection, we have:

$$\prod_{v_i \in A} (R \cap \mathfrak{p}_{v_i})^{a_i} \subseteq \bigcap_{v_i \in A} (R \cap \mathfrak{p}_{v_i}^{a_i}) = R \cap z^{-1}R \subseteq \mathfrak{p}$$

Thus, there exists some i such that $R \cap \mathfrak{p}_{v_i} \subseteq \mathfrak{p}$. Because v_i is non-trivial, $R \cap \mathfrak{p}_{v_i}$ is a non-zero prime ideal, and hence maximal by our assumption. Since $\mathfrak{p}$ is also maximal, we must have exactly:

$$\mathfrak{p} = R \cap \mathfrak{p}_{v_i}$$

(1.b) Suppose that $\Omega_{\mathfrak{p}}$ is an infinite set and for arbitrary $v \in \Omega_{\mathfrak{p}}$ we have

$$\mathfrak{p}_v \cap R_{\mathfrak{p}} = \mathfrak{p}R_{\mathfrak{p}}.$$

Since $\mathfrak{p}_v \cap R$ is a prime ideal and therefore it is maximal. $\mathfrak{p}_v \cap R_{\mathfrak{p}}$ is the prime ideal of $R_{\mathfrak{p}}$ lying over $\mathfrak{p}_v \cap R$, then we have $\mathfrak{p}_v \cap R_{\mathfrak{p}}$ is also maximal implying what we want. Therefore

$$\bigcap_{v \in \Omega_{\mathfrak{p}}} \mathfrak{p}_v \supseteq \mathfrak{p}R_{\mathfrak{p}}$$

For arbitrary nonzero element $x \in \mathfrak{p}R_{\mathfrak{p}}$ and $v \in \Omega_{\mathfrak{p}}$, we have $v(x) \neq 0$ which contradicts to (R2) since we suppose $\Omega_{\mathfrak{p}}$ is an infinite set.

(1.c) First I claim that

$$R_{\mathfrak{p}} = \bigcap_{v \in \Omega_{\mathfrak{p}}} R_v.$$

It is easy to see that $R_{\mathfrak{p}} \subseteq \bigcap_{v \in \Omega_{\mathfrak{p}}} R_v$ by definition. Conversely, suppose that $x \in \bigcap_{v \in \Omega_{\mathfrak{p}}} R_v$. Then we obtain that $v'(x) \geq 0$ for arbitrary $v' \in \Omega_{\mathfrak{p}}$. For arbitrary $v \in (\Omega \setminus \Omega_{\mathfrak{p}})$, we have $\mathfrak{p} \neq R \cap \mathfrak{p}_v$. If not, v must be contained in $\Omega_{\mathfrak{p}}$. Therefore, by (1.a), we find that $R \cap x^{-1}R$ is not contained in $\mathfrak{p}$. Suppose that $y \in (R \cap x^{-1}R) - \mathfrak{p}$, then we find for arbitrary $v \in \Omega$, $v(y), v(xy) \geq 0$ and for arbitrary $v' \in \Omega_{\mathfrak{p}}$, we have $v'(y) = 0$. It means that $xy \in R \subseteq R_{\mathfrak{p}}$ and $y \in R - \mathfrak{p}$. Since $v'(R - \mathfrak{p}) = v'((R_{v'} - \mathfrak{p}_{v'}) \cap R) = 0$ and $v'(\mathfrak{p}) = v'(\mathfrak{p}_{v'} \cap R) \geq 1$, then it must have $y \in R - \mathfrak{p}$, therefore $x \in R_{\mathfrak{p}}$ meaning that $R_{\mathfrak{p}} \supseteq \bigcap_{v \in \Omega_{\mathfrak{p}}} R_v$.

Select an arbitrary valuation $v \in \Omega_{\mathfrak{p}}$, I claim that $R_v = R_{\mathfrak{p}}$. From the discussion of (1.b), we obtain that

$$v_{\mathfrak{p}} \cap R_{\mathfrak{p}} = \mathfrak{p}R_{\mathfrak{p}}.$$

It means that $v(R_{\mathfrak{p}} - \mathfrak{p}R_{\mathfrak{p}}) = 0$. Hence, we have

$$v_{\mathfrak{p}}^2 \cap R_{\mathfrak{p}} = \mathfrak{p}^2 R_{\mathfrak{p}}.$$

We notice that $v_{\mathfrak{p}}^2 \cap R_{\mathfrak{p}} \supseteq \mathfrak{p}^2 R_{\mathfrak{p}}$ since $v(\mathfrak{p}^2 R_{\mathfrak{p}}) \geq v(\mathfrak{p}^2) \geq 2$. Therefore, $v_{\mathfrak{p}}^2 \cap R_{\mathfrak{p}}$ must equal $\mathfrak{p}^2 R_{\mathfrak{p}}$ or $\mathfrak{p}R_{\mathfrak{p}}$. If it equals $\mathfrak{p}R_{\mathfrak{p}}$, then for arbitrary $x \in K^*$, if $v(x) = 1$ then we have $x \notin R_{\mathfrak{p}}$, which is wrong. To see that, we choose an arbitrary element $x \in K^*$ such that $v(x) = 1$. Suppose that $\Omega_{\mathfrak{p}} = \{v, v_1, \dots, v_k\}$. Then we use the weak approximation

theorem, there exists an element $y \in K^*$ such that $v(y) = 0$ and $v_i(y) = -v_i(x)$. Then we find that $v(xy) = 1$ and $v_i(xy) = 0$ meaning that $xy \in R_{\mathfrak{p}}$. Therefore, it must have $v_{\mathfrak{p}}^2 \cap R_{\mathfrak{p}} = \mathfrak{p}^2 R_{\mathfrak{p}}$. Use same method, we can prove that $v_{\mathfrak{p}}^j \cap R_{\mathfrak{p}} = \mathfrak{p}^j R_{\mathfrak{p}}$ for any positive integer j . It meaning that $R_v = R_{\mathfrak{p}}$. Since select arbitrarily, the uniqueness is obtained.

(2) Let $\mathfrak{p} = \mathfrak{p}_v \cap R$. It is a prime ideal and hence is maximal ideal. We find that $v(R - \mathfrak{p}) = 0$ and therefore $v(R_{\mathfrak{p}}) \geq 0$ implying $R_{\mathfrak{p}} \subseteq R_v$. By (1), we have $R_{\mathfrak{p}} = R_v$. If there exists a different maximal ideal $\mathfrak{q}$ such that $R_v = R_{\mathfrak{q}}$. Then it means that $\mathfrak{p}_v$ is lying over $\mathfrak{q}$, then it must equal to $\mathfrak{p}$.

(3) We just need to prove that $R_{\mathfrak{p}} \subseteq R_v$ according to (1). It is quite clear from the construction of (2).

(4.a) We notice that $v_{\mathfrak{p}}(I) \neq 0$ if and only if $I \subseteq \mathfrak{p}$. If set $\{\mathfrak{p} \in \text{Spm}(R) \mid v_{\mathfrak{p}}(I) \neq 0\}$ is not finite, It means that for arbitrary element $x \in I$, there exists infinite valuations v such that $v(x) \neq 0$. Then by (R2) we must have $I = (0)$ contradicting to our condition.

(4.b) Let J denote $\prod_{i=1}^n \mathfrak{p}^{r_{\mathfrak{p}_i}}$ such that for $\mathfrak{p} \notin \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$ we have $v_{\mathfrak{p}}(I) = 0$ and $r_{\mathfrak{p}} = v_{\mathfrak{p}}(I)$. We notice that $I_{\mathfrak{p}} = \mathfrak{p}^{r_{\mathfrak{p}}} R_{\mathfrak{p}}$ since

$$I_{\mathfrak{p}} \subseteq \mathfrak{p}^{r_{\mathfrak{p}}} R_{\mathfrak{p}} \text{ and } I_{\mathfrak{p}} \subsetneq \mathfrak{p}^{r_{\mathfrak{p}}+1} R_{\mathfrak{p}}$$

Then we have

$$I = \bigcap_{\mathfrak{p} \in \text{Spm}(R)} I_{\mathfrak{p}} = \bigcap_{i=1}^n I_{\mathfrak{p}_i} = J.$$

The last step is because $\mathfrak{p}_i$ are pairwise coprime. (4.c) It is directly deduced by (4.b).

(5) Noetherian is due to (4.c). For integrality, we consider ideal (x) generated by an arbitrary element $x \in R$. ■

Homework 3

Exercise 3.1

Let L be an R -lattice contained in the K -vector space V . Prove that L is an R -lattice supported in V if and only if FL has a K -free basis.

Solution: Suppose that L is supported in V . It means that $\dim_F(FL) = \dim_K(KL)$. Choose a basis $\{e_i\}$ of FL . Since $\{e_i\}$ spans KL , then if $\{e_i\}$ is not a basis of KL , we then have $\dim_K(KL) < \dim_F(FL)$. This is a contradiction. Conversely, it is trivial. ■

Homework 4

Exercise 4.1

Let $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ be prime ideals of a ring R . Prove that for any ideal $I \subseteq R$, $I \subseteq \bigcup_{i=1}^m \mathfrak{p}_i$ if and only if $I \subseteq \mathfrak{p}_i$ for some $1 \leq i \leq m$.

Solution: Suppose that $I \subseteq \bigcup_{i=1}^m \mathfrak{p}_i$. We use induction on n and starting by $n = 2$. Suppose that $a_1 \in \mathfrak{p}_1 \cap \mathfrak{p}_2^c \cap I$ and $a_2 \in \mathfrak{p}_1^c \cap \mathfrak{p}_2 \cap I$ where both a_1 and a_2 are not equal to zero. Since I is an ideal, we then have $a_1 + a_2 \in I$. Clearly, $a_1 + a_2$ must not be contained in $\mathfrak{p}_1$ and $\mathfrak{p}_2$. Otherwise, we then have $a_1 \in \mathfrak{p}_2$ or $a_2 \in \mathfrak{p}_1$. It is a contradiction.

Suppose our claim holds for $n - 1$. Suppose that there exist an element $a_i \in I \cap \mathfrak{p}_i$ such that a_i is not contained in any $I \cap \mathfrak{p}_j$ for $j \neq i$. Otherwise, we can reduce it to $n - 1$ case. Then we consider the elements $a = \sum_i \prod_{k \neq i} a_k$. Since I is an ideal then we have $a \in I$. Suppose that $a \in \mathfrak{p}_j \cap I$. Then we obtain that $\prod_{k \neq j} a_k \in \mathfrak{p}_j$. Since $\mathfrak{p}_j$ is a prime ideal and $a_i \notin \mathfrak{p}_j$ for $i \neq j$, we then have $\prod_{k \neq j} a_k \notin \mathfrak{p}_j$. It is a contradiction. Therefore, our claim holds. ■

Exercise 4.2

Let R be a Dedekind domain, and let $S \subseteq R \setminus \{0\}$ be a multiplicative subset.

1. Prove that the ring of fractions $S^{-1}R$ is a Dedekind domain.
2. Suppose that $S = R \setminus \bigcup_{i=1}^m \mathfrak{p}_i$ for finitely many maximal ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ of R . Prove that $S^{-1}R$ is a PID.

Solution: (1) For each $\mathfrak{p} \cap S = \emptyset$, we have

$$(S^{-1}R)_{S^{-1}\mathfrak{p}} = S^{-1}R \otimes_R (R \setminus \mathfrak{p})^{-1} = S^{-1}R_{\mathfrak{p}} = R_{\mathfrak{p}}$$

Since R is Dedekind domain, we then have $R_{\mathfrak{p}}$ is DVR. Therefore, $S^{-1}R$ is also a Dedekind domain.

(2) Since Dedekind domain $S^{-1}R$ only have finite prime ideal $S^{-1}\mathfrak{p}_i$. According to Exercise 1.2, it is a PID. ■

Exercise 4.3

Let R be a Dedekind domain, and let $M \neq 0$ be a finitely generated R -module. Let M_{tors} be the torsion submodule of M . The invariant factors of M are defined as those of M_{tors} (cf. Thm. 1.2.27).

1. Prove that there is a torsion-free submodule $N \subseteq M$ such that $M = M_{\text{tors}} \oplus N$, and that the isomorphism class of N is uniquely determined by M .

We define the **dimension** of M and the **Steinitz class** of M as the dimension of N and the Steinitz class of N respectively.

2. Prove that the isomorphism class of M is determined by its invariant factors, dimension and Steinitz class.

Solution: (1) Because M/M_{tors} is a R -lattice. It is projective since R is Dedekind domain. We have the exact sequence

$$0 \rightarrow M_{\text{tors}} \rightarrow M \rightarrow M/M_{\text{tors}} \rightarrow 0$$

split. We then have $M = M/M_{\text{tors}} \oplus M_{\text{tors}}$.

(2) Let $M \simeq M'$. We have $M_{\text{tors}} \simeq M'_{\text{tors}}$ and $M/M_{\text{tors}} \simeq M'/M'_{\text{tors}}$. The torsion-free part is uniquely determined by its Steinitz class and dimension. According to our textbook, the invariant factors are uniquely determined by torsion part M_{tors} . We just need to prove that for integral ideal $\mathfrak{a}_i$ and $\mathfrak{b}_j$ such that $\mathfrak{a}_i \supseteq \mathfrak{a}_{i+1}$ and $\mathfrak{b}_j \supseteq \mathfrak{b}_{j+1}$ if

$$\bigoplus_{i=1}^n R/\mathfrak{a}_i \simeq \bigoplus_{j=1}^m R/\mathfrak{b}_j$$

we then have $\mathfrak{a}_i = \mathfrak{b}_i$. It is also trivial, since for every prime ideal $\mathfrak{q}$ of R , this two module must agree on those localization.

If $\mathfrak{q} \supseteq \mathfrak{a}_n$ but $\mathfrak{q} \not\supseteq \mathfrak{b}_m$, then we have

$$\left(\bigoplus_{j=1}^m R/\mathfrak{b}_j \right)_{\mathfrak{q}}$$

is a zero $R_{\mathfrak{q}}$ -module. However,

$$\left(\bigoplus_{i=1}^n R/\mathfrak{a}_i \right)_{\mathfrak{q}}$$

is a torsion $R_{\mathfrak{q}}$ -module. It is impossible. Therefore, we must have $\mathfrak{a}_n$ and $\mathfrak{b}_m$ shares the same prime factors. Moreover, since $R_{\mathfrak{q}}$ is a DVR, and thus is a PID. According to the classification of modules over PID and the fact that the prime elements on DVR are exactly uniformizers of $R_{\mathfrak{q}}$. It implies that

$$(\mathfrak{a}_i)_{\mathfrak{q}} = (\mathfrak{b}_i)_{\mathfrak{q}}.$$

and $n = m$. Our claim holds. ■

Exercise 4.4

Let R be an integral domain with fraction field F . Let L be a nonzero R -lattice in a finite dimensional F -vector space V . Prove that $\dim L$ is equal to the largest positive integer n such that L contains a linearly independent system of n vectors of V .

Solution: Suppose that $\omega_1, \dots, \omega_n$ is such a system. Suppose that $\omega \in FL$ can not be linearly represented by ω_i over V . Therefore, $\{\omega, \omega_1, \dots, \omega_n\}$ is linearly independent over F . Since R is integral domain and F is the fraction field of R , there exist an element $r \in R$ such that $\{r\omega, r\omega_1, \dots, r\omega_n\}$ is included in L and for each of those elements is not zero. Then we find a linearly independent system larger than previous one, which is a contradiction. ■

Exercise 4.5

Let R be a Dedekind integral domain with fraction field F . Let L be a nonzero R -lattice in a finite dimensional F -vector space V .

Prove that for any nonzero $x \in L$ the following are equivalent:

1. x is a primitive vector in L (in the sense of Definition 1.2.17).
2. the quotient R -module L/Rx is torsion-free.
3. x is a maximal vector in L (in the sense of (1.2.31)).

Solution: We first show that (1) implies (2). x is primitive means the exact sequence

$$0 \rightarrow Rx \rightarrow L \rightarrow L/Rx \rightarrow 0$$

splits and $L \simeq Rx \oplus L/Rx$. If L/Rx is torsion-free, then L has a torsion element, then L a torsion element. Consequently, it cannot be a R -lattice.

(2) also implies (1). Since L/Rx as a R -lattice, it is projective R -module, and thus the exact sequence splits.

Next, we show that (2) implies (3). Let $\mathfrak{a}$ be the coefficient ideal of x . Then we have $\mathfrak{a}x/Rx$ is torsion-free module. It also say that

$$\mathfrak{a}/R \simeq R/\mathfrak{a}^{-1}$$

is a torsion-free module. Actually, we choose an arbitrary element d in determinant ideal $\det(\mathfrak{a})$, we then have $d\mathfrak{a} \subseteq R$. It holds if and only if $\mathfrak{a} = R$. Therefore, x is maximal.

We check (3) implies (2). Let $\mathfrak{a}$ be the coefficient ideal of x . Suppose that $L/\mathfrak{a}x$ has torsion element $y + \mathfrak{a}x$. It means that there exists r such that $ry \in \mathfrak{a}x$ but $y \notin \mathfrak{a}x$. We can suppose that $ry = ax$ and $a/rx \notin \mathfrak{a}x$ means $a/r \notin \mathfrak{a}$. Then we have $y = a/rx \notin L$. It is a contradiction.

Therefore, statements are all equivalent. ■

Homework 5

Exercise 5.1

Let I, J be fractional ideals of a Dedekind domain R . Prove that $I^{-1} \cap J^{-1} = (I + J)^{-1}$.

Solution: Let $I^{-1} = \mathfrak{p}_1^{-a_1} \cdots \mathfrak{p}_n^{-a_n}$ and $J^{-1} = \mathfrak{p}_1^{-b_1} \cdots \mathfrak{p}_n^{-b_n}$. Then $I + J = \mathfrak{p}_1^{c_1} \cdots \mathfrak{p}_n^{c_n}$ where $c_i = \min\{a_i, b_i\}$. Let v_i be the valuations correspond to $\mathfrak{p}_i$ respectively. Then we have

$$\begin{aligned} I^{-1} \cap J^{-1} &= \{r \in F \mid v_i(r) \geq \max\{-a_i, -b_i\} \text{ and } w(r) \geq 0 \text{ for others valuation } w\} \\ &= \{r \in F \mid v_i(r) \geq -\min\{a_i, b_i\} \text{ and } w(r) \geq 0 \text{ for others valuation } w\} \\ &= (I + J)^{-1}. \end{aligned}$$
■

Exercise 5.2

Let J be an integral ideal of a Dedekind domain R . Prove that for every fractional ideal I of R , one has $I/IJ \cong R/J$ as R -modules.

Solution: Let $\mathfrak{p}$ be an arbitrary prime ideal. Then we have isomorphism of $R_{\mathfrak{p}}/J_{\mathfrak{p}}$ -module

$$(I/IJ)_{\mathfrak{p}} = I_{\mathfrak{p}}/(I_{\mathfrak{p}}J_{\mathfrak{p}}) \simeq R_{\mathfrak{p}}/J_{\mathfrak{p}}.$$

The isomorphism is given by $R_{\mathfrak{p}} \simeq I_{\mathfrak{p}}$. Interpret I/IJ and R/J be the R/J -module. According to Structure Theorem for Artinian Modules, we have

$$I/IJ \cong R/J.$$

Hence it is an isomorphism as R -modules. ■

Exercise 5.3

Prove that the existence statement in Thm.1.2.27 can be deduced from Thm.1.2.39.

Solution: Let $\{x_1, \dots, x_n\}$ be the generators of torsion R -module M . We consider the homomorphism

$$\begin{aligned} R^n &\rightarrow M \\ (r_i)_i &\mapsto \sum r_i x_i \end{aligned}$$

which is a epimorphism. Suppose its kernel is N . Clearly, N is a torsion-free module with finite generators since R is Noetherian. Therefore N is a projective R -module. Consider the short exact sequence

$$0 \rightarrow N \rightarrow R^n \rightarrow M \rightarrow 0$$

It implies that

$$K^n \simeq N \otimes_R K.$$

Since functor $\otimes_R K$ is exact and $M \otimes_R K$ is zero.

Then there is a basis such that

$$\begin{aligned} R^n &= \mathfrak{a}_1 y_1 + \mathfrak{a}_2 y_2 + \dots + \mathfrak{a}_n y_n \\ N &= \mathfrak{a}_1 \mathfrak{b}_1 y_1 + \mathfrak{a}_2 \mathfrak{b}_2 y_2 + \dots + \mathfrak{a}_n \mathfrak{b}_n y_n \end{aligned}$$

and $\mathfrak{b}_i \supseteq \mathfrak{b}_{i+1}$. Then we obtain that

$$R^n/N \simeq \bigoplus_i \mathfrak{a}_i/(\mathfrak{a}_i \mathfrak{b}_i) \simeq \bigoplus_i R/\mathfrak{b}_i.$$

■

Exercise 5.4

Prove the following facts about metabolic bilinear spaces over a field F :

1. The hyperbolic spaces $m.\mathbb{H}$ are metabolic.
2. For any symmetric bilinear spaces V, V' , if two of the three spaces V, V' and $V \perp V'$ are metabolic, then so is the third. In particular, an orthogonal sum of metabolic spaces is metabolic.

3. For any nonsingular symmetric bilinear space (V, b) , the space $(V, b) \perp (V, -b)$ is metabolic.
4. Let V be a 2-dimensional nonsingular symmetric space. Then V is metabolic if and only if it is isotropic.
5. Every 2-dimensional metabolic space is isomorphic to $\mathbb{H}$ or $\langle a, -a \rangle_{\text{bil}}$ for some $a \neq 0$.
6. If $\text{char}(F) \neq 2$, then every metabolic bilinear space is hyperbolic.

Solution: (1) Clearly, $m\mathbb{H}$ it is not singular of even dimension. Let $\{x_i, y_i\}$ be the basis of i^{th} hyperbolic subspace of $m\mathbb{H}$, then we let $V = \text{span}(x_i)$. It is easy to check that V is totally isotropic and $\dim(V) = m$.

(2) Suppose that V and V' are metabolic. Clearly, $V \perp V'$ is not singular of even dimension. Let W and W' be the totally isotropic subspace of V and V' respectively and hence have half dimension. Then $W \perp W'$ also be the totally isotropic subspace of $V \perp V'$ with a half of dimension. Therefore, $V \perp V'$ is totally isotropic.

Suppose that V and $V \perp V'$ is totally isotropic. Then we have $\dim(V') = \dim(V \perp V') - \dim(V)$ is even. And V' must be not singular. Otherwise, $V \perp V'$ will be singular. Let W be the totally isotropic subspace of $V \perp V'$ with a half dimension. Then we have $W = (W \cap V) \perp (W \cap V')$. I claim that $\dim(W \cap V) \leq \dim(V)/2$. If $\dim(W \cap V) > \dim(V)/2$, then we choose a basis of $W \cap V$ and extend it to be a basis of V . Let we denote it by $\{e_1, \dots, e_n, \dots, e_{2n}\}$. Write the Gram matrix over this basis and using Laplace expansion, we deduce that V is singular. It is a contradiction. Therefore, we have $\dim(V' \cap W) \geq \dim(W) - \dim(V)/2 = \dim(V')/2$.

(3) Let e_i be a standard basis of (V, b) . Then it is also a standard basis of $(V, -b)$. Consider the basis $(e_i, e_i), (e_i, -e_i)$ of $V \oplus V$. Check subspace generated by (e_i, e_i) is totally isotropic subspace with half dimension of $(V, b) \perp (V, -b)$.

(4) If it is isotropic, then it has Gram matrix

$$\begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}$$

with $a \in F^*$. It is a hyperbolic and thus is a metabolic. Conversely, if it is a metabolic, it has the Gram matrix

$$\begin{pmatrix} 0 & a \\ a & c \end{pmatrix}$$

It implies that V is isotropic.

(5) If $c = 0$, then V is isomorphic to $\mathbb{H}$. If $c \neq 0$, let $\{e_1, e_2\}$ be the basis corresponding to this matrix. Then we consider the basis $\{e_2, e_2 - a^{-1}ce_1\}$. It has the Gram matrix

$$\begin{pmatrix} c & 0 \\ 0 & -c \end{pmatrix}$$

The statements holds.

(6) Let U be the totally isotropic subspace of V with half dimension. According to our textbook, metabolic space can be decomposit to $V_1 \perp \dots \perp V_n$ where $V_i = Fx_i \otimes Fy_i$ is a subspace of dimension 2 where $(x_i, y_i) = 1$ and $x_i \in U$. Therefore, V_i is isotropic.

Suppose that V is isotropic to the second case of (5). We consider basis $\{x_i + y_i, x_i - y_i\}$. It has the Gram matrix

$$\begin{pmatrix} 0 & 2c \\ 2c & 0 \end{pmatrix}$$

Since $\text{char}(F) \neq 2$, then we have V_i is hyperbolic space and $V \simeq n\mathbb{H}$. ■

Homework 6

In the following exercises, F denotes a field of characteristic $\text{char}(F) \neq 2$. In principle, the following exercises can be done by using only results stated in §2.1.

Exercise 6.1

Let V be a nonsingular quadratic space over F . Prove that V is isotropic if and only if V represents $\mathbb{H}$.

Solution: Suppose that $Q(x) = 0$. Since V is not singular, then there exists $y \in V$ such that $V = V' \perp (Fx \oplus Fy)$. Since characteristic is not equal 2, then we have isotropic subspace $Fx \oplus Fy$ is hyperbolic. Therefore we have representation $\varphi : \mathbb{H} \rightarrow Fx \oplus Fy$. Converse is trivial. ■

Exercise 6.2

Let V be a quadratic space over F and $\alpha \in F^\times$. Prove that the following statements are equivalent:

- (i) V represents the 1-dimensional space $\langle \alpha \rangle$.
- (ii) There is a splitting $V = \langle \alpha \rangle \perp W$.

Prove that if V is nonsingular, then the above conditions are also equivalent to the following:

- (iii) The quadratic space $V \perp \langle -\alpha \rangle$ is isotropic.

Solution: We show that (1) implies (2). Since $\alpha \in F^*$, then $\langle \alpha \rangle$ is not singular. Then according to the textbook, we have $V \simeq \langle \alpha \rangle \perp W$. Converse is trivial.

Suppose that V is nonsingular. Then clearly (ii) implies (iii) by considering the $\langle \alpha \rangle \perp \langle -\alpha \rangle$. Conversely, $V \perp \langle -\alpha \rangle$ is also nonsingular. Then according to Exercise 6.1, we have $V \perp \langle -\alpha \rangle$ represents a hyperbolic space $W \simeq \mathbb{H}$ such that $V \perp \langle -\alpha \rangle = W \perp W^\perp$. Since $\text{char}(F) \neq 2$, W represents

$$\begin{pmatrix} \alpha & 0 \\ 0 & -\alpha \end{pmatrix}$$

Using Witt cancellation then we have $V \simeq \langle \alpha \rangle \perp W^\perp$. Therefore, V represents the 1-dimensional space $\langle \alpha \rangle$. ■

Exercise 6.3

Prove that Witt cancellation holds for all quadratic spaces in characteristic $\neq 2$. More precisely, for any (possibly singular) quadratic spaces V, V_1, V_2 over F , prove that $V \perp V_1 \cong V \perp V_2$ if and only if $V_1 \cong V_2$.

Solution: The direction $(\Leftarrow)$ is trivial. We prove $(\Rightarrow)$.

Suppose there is an isomorphism $V \perp V_1 \cong V \perp V_2$.

Since $\text{char}(F) \neq 2$, the relation $B(x, x) = 2Q(x)$ guarantees that the radical of any quadratic space is a totally singular space. This ensures the existence of a valid orthogonal decomposition for any space U :

$$U = \text{rad}(U) \perp U_{nd}$$

where U_{nd} is a strictly nonsingular subspace.

An isometry must preserve the radical. Applying this to our spaces gives:

$$\text{rad}(V) \perp \text{rad}(V_1) \cong \text{rad}(V) \perp \text{rad}(V_2)$$

For totally singular spaces, isometry is entirely determined by dimension. Taking dimensions on both sides and canceling $\dim(\text{rad}(V))$ yields $\dim(\text{rad}(V_1)) = \dim(\text{rad}(V_2))$. Thus:

$$\text{rad}(V_1) \cong \text{rad}(V_2)$$

Furthermore, the isometry also preserves the nonsingular orthogonal complements:

$$V_{nd} \perp (V_1)_{nd} \cong V_{nd} \perp (V_2)_{nd}$$

Because all components here are strictly nonsingular and $\text{char}(F) \neq 2$, we are permitted to invoke the standard Witt's Cancellation Theorem (which relies on non-degenerate reflections) to cancel V_{nd} from both sides. This yields:

$$(V_1)_{nd} \cong (V_2)_{nd}$$

Finally, we reconstruct V_1 and V_2 by orthogonally summing their respective isometric parts:

$$V_1 = \text{rad}(V_1) \perp (V_1)_{nd} \cong \text{rad}(V_2) \perp (V_2)_{nd} = V_2$$

This completes the proof. ■

Homework 7

Exercise 7.1

Let $(V, \langle \cdot, \cdot \rangle)$ be a nonsingular bilinear space and let $x_1, \dots, x_n$ be a basis of V . Let $y_1, \dots, y_n \in V$ be the dual basis of $x_1, \dots, x_n$ with respect to the given bilinear form $\langle \cdot, \cdot \rangle$. Prove

$$\det(\langle x_i, x_j \rangle) \det(\langle y_i, y_j \rangle) = 1.$$

Solution: Suppose that $(x_i) = A(y_i)$. Then we have $x_i = \sum_j a_{ij}y_j$. Since (y_i) is the dual basis of (x_i) , then we have

$$\langle x_i, x_j \rangle = \left\langle \sum_k a_{ik}y_k, x_j \right\rangle = a_{ij}.$$

Next, we consider $A^*(x_i) = \det(A)(y_i)$. We let $A^* = (a_{ij}^*)$.

$$\det(A)\langle y_i, y_j \rangle = \langle y_i, \sum_k a_{jk}^*x_k \rangle = a_{ji}^*.$$

We find that

$$\det(\langle x_i, x_j \rangle) \det(\langle y_i, y_j \rangle) = \det(A) \det(A^*) \det(A^{-n}) = 1$$

■

Exercise 7.2

Consider the quadratic $\mathbb{Z}$ -lattice (L, Q) with $L = \mathbb{Z}^{\oplus 2}$ and $Q(x, y) = 2xy$. Prove that L cannot be decomposed as an orthogonal sum of two 1-dimensional lattices.

Solution: Let e_1, e_2 denotes $(1, 0)$ and $(0, 1)$ respectively. Then we have

$$\langle ae_1 + be_2, ce_1 + de_2 \rangle = 2(bc + ad).$$

If $ae_1 + be_2, ce_1 + de_2$ are orthogonal, then we must have $bc + ad = 0$. Hence, $ae_1 + be_2, ce_1 + de_2$ must be the generators of L , which means the relative determinant must be ± 1 , which is $ad - bc = \pm 1$. Then we have

$$2bc = \mp 1.$$

It is impossible, since we must have $b, c \in \mathbb{Z}$.

■

Homework 9

Exercise 9.1

Let (V, Q) be a quadratic space over a field F of characteristic $\neq 2$.

1. Check that for any anisotropic vector $x \in V$, the symmetry τ_x defined in (2.3.1) is an isometry of (V, Q) .
2. Assume that (V, Q) is nonsingular. Without using the Cartan–Dieudonné theorem, prove that $\det(\tau_x) = -1$ for any anisotropic vector $x \in V$ and that $\det(\sigma) = \pm 1$ for every $\sigma \in O(V)$.

Solution: (1) We let $B(x, y) = Q(x + y) - Q(x) - Q(y)$. Clearly, τ_x is invertible since $\tau_x^2 = \text{id}$. On one hand, we have

$$B\left(y, \frac{-B(x, y)}{Q(x)}x\right) = Q(\tau_x(y)) - Q(y) - \frac{B^2(x, y)}{Q^2(x)}Q(x).$$

On the other hand we have

$$B\left(y, \frac{-B(x, y)}{Q(x)}x\right) = -\frac{B^2(x, y)}{Q(x)}.$$

Then we obtain that $Q(\tau_x(y)) = Q(y)$

(2) Since V is not singular, there exists an orthogonal decomposition $V = \langle x \rangle + \langle x \rangle^\perp$. Therefore, the linear transformation τ_x gives matrix of diagonal form $\text{diag}(-1, 1, \dots, 1)$. Then we obtain that $\det(\tau_x) = -1$.

To prove that $\det(\sigma) = \pm 1$. Take $\{e_i\}$ be the orthogonal basis whose Gram matrix is given by G . Let M be the matrix representing the linear transformation σ . Then we have

$$M^T G M = G$$

and hence $\det(G) \det^2(M) = \det(G)$. We then find that $\det(M) = \pm 1$. ■

Exercise 9.2

Let M be a subgroup of the additive group $\mathbb{R}$ endowed with the usual Euclidean topology. Prove that the following assertions are equivalent:

- (i) M is not dense in $\mathbb{R}$.
- (ii) There exists a real number $\varepsilon > 0$ such that $M \cap]-\varepsilon, \varepsilon[= \{0\}$.
(Here $]-\varepsilon, \varepsilon[$ denotes the open interval $\{x \in \mathbb{R} \mid -\varepsilon < x < \varepsilon\}$.)
- (iii) M is discrete in $\mathbb{R}$, i.e., the subspace topology on M is the discrete topology.
- (iv) $M = r\mathbb{Z}$ for some $r \in \mathbb{R}$.

Solution: We first show that (1) implies (2). Suppose that M is dense on 0. Then for arbitrary $x \in M$ and $y \in \mathbb{R}$, there exists $\varepsilon \in M$ such that $|x - y| < \varepsilon$. If $x > y$, it implies that $x > x - \varepsilon > y$ and if $x < y$, it implies that $x < x + \varepsilon < y$. In any case, we find an element $x \pm \varepsilon$ is closer than x to y . Equivalently we write $d(M, y) = 0$. It contradicts to (1).

Clearly, (2) also implies (1) since M is not dense on zero.

Hence, (2) implies (3). Because M must be nowhere dense. If M is dense on x , for arbitrary $\varepsilon > 0$ such that sufficiently small, then there exists an $x + \eta_1, x + \eta_2 \in M$ such that $|\eta_1 - \eta_2| < \varepsilon$. It follows that M is dense on zero, which contradicts to (2).

(3) implies (2) also holds by definition.

(4) implies (2) is trivial.

Finally, we check that (3) implies (4). We consider the set of distance between elements in M , denoted by

$$D = \{d(x, y) \mid x, y \in M\}.$$

Since M is discrete, we can find a minimal value δ of D . Clearly, we have $\delta\mathbb{Z} \subseteq M$. If there exists an element $y \in M$ is not contained in $\delta\mathbb{Z}$, then we find that $d(y, \delta\mathbb{Z}) < \delta$. It contradicts to the minimality of δ . Therefore, we then have $M = \delta\mathbb{Z}$. ■

Exercise 9.3

Let K be a field and let v be an exponential valuation on K with valuation ring A . Let $(\hat{K}, \hat{v})$ be the completion of the valued field (K, v) .

1. Let $\hat{A}$ denote the valuation ring of $\hat{v}$ in $\hat{K}$. Prove that $\hat{A}$ equals the topological closure of A in $\hat{K}$.
2. Show that $\hat{v}(\hat{K}^\times) = v(K^\times)$ and that the natural map $A \rightarrow \hat{A}$ induces an isomorphism on their residue fields. In particular, if v is discrete (resp. discrete and normalized), then so is $\hat{v}$.

Solution: Maybe we should show stability first.

(1) Let $\bar{A}$ denote the topological closure of A . Let $||$ denote the absolute valuate determined by v . Let (x_i) be a Cauchy sequence over K . Suppose that $(x_i) \in \hat{A}$, it means that $\lim_{n \rightarrow \infty} |x_n| \leq 1$. Then it is to said that there exists an integer $N > 0$ for any $n > N$ we have $x_n \in A$. Then we consider constant sequences $(x_{N+k})_i$. For arbitrary $\varepsilon > 0$, there exists an integer N such that for $n > N$ we have

$$|x_n - x_{N+k}| < \varepsilon$$

for arbitrary $k > 0$. It equivalently say given any $\varepsilon > 0$, we find an element $(x_{N+k})_i \in A$ such that $|(x_{N+k})_i - (x_i)_i| < \varepsilon$ where $|||$ denotes the absolute valuate induced by $\hat{v}$. Thus we have $\hat{A} \subseteq \bar{A}$.

Conversely, since $\hat{A}$ is closed in $\hat{K}$ and it contains A , we then have $\bar{A} \subseteq \hat{A}$. Therefore, we obtain that $\bar{A} = \hat{A}$.

(2) We just need to show that any Cauchy sequence (x_i) admits a stable sequence of valuation $v(x_i)$ if (x_i) is not the zero over $\hat{K}$. To see that we use the strongly triangular inequality. Let $a = \lim_{n \rightarrow \infty} |x_n|$. There exists $n > 0$ such that for any k we have

$$|x_n - x_{n+k}| < a/2.$$

We can also choose sufficiently large of n such that

$$||x_{n+k}| - a| < a/2$$

In this sense we have $|x_{n+k}| > a/2 > |x_n - x_{n+k}|$, then

$$|x_n| = |x_n - x_{n+k} + x_{n+k}| = |x_{n+k}|.$$

Therefore we show that $|x_n|$ is stable and thus $\hat{v}(\hat{K}^\times) = v(K^\times)$.

Let $\mathfrak{m}$ and $\hat{\mathfrak{m}}$ denote the maximal prime ideal of A and $\hat{A}$ respectively. Consider the homomorphism

$$A \rightarrow \hat{A}/\hat{\mathfrak{m}}.$$

It is epimorphic. For arbitrary Cauchy sequence (x_i) such that $\lim_{n \rightarrow \infty} |x_n| = 1$, we suppose it is stable at k^{th} . Hence we can suppose that $|x_{k+n} - x_k| < 1/2$, then we find that $(x_n - x_k)_n \in \hat{\mathfrak{m}}$ and therefore (x_n) have preimage $(x_k)_n$. Its kernel is $A \cap \hat{\mathfrak{m}} = \mathfrak{m}$.

Our claim holds. ■

Exercise 9.4

Let R be a proper subring of a field K such that for every $x \in K^\times$, either $x \in R$ or $x^{-1} \in R$. (We say that R is a **valuation ring** in K .)

1. Prove that R is integrally closed.
2. Let v be a nontrivial exponential valuation on a field K . Show that the ring $\mathcal{O}_v = \{x \in K \mid v(x) \geq 0\}$ is a valuation ring in the above sense, i.e., for all $x \in K^\times$, either $x \in \mathcal{O}_v$ or $x^{-1} \in \mathcal{O}_v$. Deduce that $\mathcal{O}_v$ is integrally closed.

Solution: (1) Suppose that it is not integrally closed. Equivalently, there exists an integral element $x \in K \setminus R$. Suppose that

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$$

Then we have

$$a_0x^{-n} + a_1x^{-n+1} + \cdots + 1 = 0$$

Hence we have

$$x = -(a_0x^{-n+1} + a_1x^{-n+2} + \cdots + a_{n-1}).$$

It implies that $x \in R$.

(2) If $x \notin \mathcal{O}_v$, then we have $v(x) < 0$. Therefore, we must have $v(x^{-1}) > 0$ and $x^{-1} \in \mathcal{O}_v$. The integrality of $\mathcal{O}_v$ follows. ■

Exercise 9.5

Let $|\cdot|_1, \dots, |\cdot|_n$ be pairwise inequivalent non-archimedean absolute values of a field K . Prove that for all $a_1, \dots, a_n \in K^\times$ there exists an element $x \in K^\times$ such that $|x|_i = |a_i|_i$ for each $i = 1, \dots, n$.

Solution: We use Weak Approximation Theorem. Let $\varepsilon < \min_i \{|a_i|_i\}$, then there exist an element $x \in K$ such that

$$|x - a_i|_i < \varepsilon.$$

Next, we consider

$$|x|_i = |x - a_i + a_i|_i = |a_i|_i$$

since $|x - a_i|_i < |a_i|_i$. ■

Exercise 9.6

Let $(K, |\cdot|)$ be a complete non-archimedean valued field. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence in K . Prove that the series $\sum_{n=0}^{\infty} a_n$ converges in K if and only if $\lim_{n \rightarrow \infty} a_n = 0$ in K .

Solution: Suppose that $\sum a_n$ converges. Then $|a_{n+k+1} - a_{n+k}| < \varepsilon$ therefore $|a_{n+k} - a_n| = |a_{n+k} - a_{n+k-1} + \cdots + a_{n+1} - a_n| < \varepsilon$ which implies that a_n converges.

Conversely, $|a_n + \cdots + a_{n+k}| \leq \max_k \{|a_{n+k}|\} < \varepsilon$. $\sum a_n$ converges. ■

Exercise 9.7

Let $(K, |\cdot|)$ be a henselian (non-archimedean valued) field.

1. Let $f(T) = a_2T^2 + a_1T + a_0$ be a polynomial of degree 2 over K . Prove that if f is irreducible over K , then $|a_1|^2 \leq |a_0| \cdot |a_2|$.
2. Let (V, Q) be an anisotropic quadratic space over K . Prove that

$$|\langle x, y \rangle_Q|^2 \leq |Q(x)| \cdot |Q(y)| \quad \text{for all } x, y \in V.$$

(Here $\langle x, y \rangle_Q := Q(x + y) - Q(x) - Q(y)$.)

Solution: (1) Suppose that $|a_1|^2 > |a_0a_2|$, it equivalently say that $a_0a_1^{-2}a_2 \in \mathfrak{p}$ where R is the valuation ring induced by absolute value. Without loss of generality, let $f(T)$ be the primitive polynomial. Let $T = a_0a_1^{-1}X$, we then obtain that

$$f(X) = a_0 \left(\frac{a_2a_0}{a_1^2} X^2 + X + 1 \right)$$

Let

$$g(X) = \frac{a_2a_0}{a_1^2} X^2 + X + 1$$

which is the polynomial with coefficients in R . Consider that

$$\bar{g}(X) = X + 1 \pmod{\mathfrak{p}}$$

Then $\bar{g}(X)$ has a simple root since $\bar{g}'(X) = 1$. Then by Hensel's Lemma, it can be lifted to R . However, $g(X)$ is irreducible and thus it can not have a root.

(2) Let $a_2 = Q(x)$, $a_0 = Q(y)$ and $a_1 = \langle x, y \rangle_Q$. Then $Q(Tx + y) = a_2T^2 + a_1T + a_0$. Since V is anisotropic, then $Q(Tx + y)$ is irreducible over K . Then according to (1) is what we want. ■

Homework 10

Exercise 10.1

Let $K = \mathbb{R}$ or $K = \mathbb{C}$, and let $|\cdot|$ be the usual absolute value on K . Let s be a positive real number. Prove that $|\cdot|^s$ is an absolute value on K if and only if $0 < s \leq 1$.

Solution: AV1 and AV2 is quite clear. We just show AV3. Suppose that $|\cdot|^s$ is an absolute value, we consider that

$$2 \geq 2^s$$

It means that $0 < s \leq 1$. Suppose that $0 < s \leq 1$, we just need to prove that

$$1 + x^s \geq (1 + x)^s$$

for $0 < x < 1$. Consider function $g(x) = 1 + x^s - (1 + x)^s$ with

$$g'(x) = sx^{s-1} - s(1 + x)^{s-1} > 0$$

Therefore $g(x) > g(0) = 0$. Our claim holds. ■